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Vitreosense: Deep-Learning Retinal Screening from Fundus Images

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Abstract: Disease detection in the eyes is still difficult because of class imbalance, small datasets, and requirements for real-time efficient systems. This work proposes OcularNet, a deep learning model that employs transfer learning using MobileNetV2 for multi-disease classification from retinal fundus images in the ODIR-5K dataset. The method fine-tunes pre-trained models to compensate for data limitations and utilizes the lightweight architecture of MobileNetV2 for efficiency. The innovation of this approach is in combining the computational lightness of MobileNetV2 with multi-label classification for diabetic retinopathy, glaucoma, and cataract. The model has benefits like decreased computation load, real-time deployment, and strong multi-disease recognition. Performance evaluation shows high accuracy (90.8%), precision (0.87), recall (0.88), F1-score (0.87), and AUC-ROC (0.92) across many diseases, handling the class imbalance problem efficiently. This method is extremely valuable in practical clinical settings, particularly in remote or underserved areas, with an automated, cost-effective, and accurate diagnosis for ocular diseases.

Keywords: Ocular Disease Detection, Deep Learning, MobileNetV2, Multi-label Classification, ODIR-5K Dataset, Transfer Learning, Diabetic Retinopathy, Glaucoma, Retina Image Analysis, Automated Diagnosis.

I. Introduction

Retinal diseases such as diabetic retinopathy, glaucoma, age-related macular degeneration, and hypertension-induced retinopathy are among the leading causes of preventable blindness worldwide, necessitating early and accurate diagnosis through retinal fundus imaging. However, the growing patient burden and limited availability of expert ophthalmologists highlight the need for automated, reliable, and clinically interpretable diagnostic systems. In this work, we propose RetinAI, a hybrid artificial intelligence-based framework for multi-label retinal disease detection that integrates deep learning with classical clinical reasoning. The system employs a DenseNet121-based convolutional neural network for simultaneous prediction of multiple retinal conditions from a single fundus image, complemented by image quality assessment to ensure reliable inputs. To enhance clinical relevance and interpretability, the model incorporates classical computer vision techniques for extracting diagnostic features such as exudates, hemorrhages, and cup-to-disc ratio, which are further validated through a rule-based engine grounded in established ophthalmological guidelines. Additionally, Grad-CAM-based visual explanations are generated to highlight regions influencing model predictions, improving transparency and trust. A unique aspect of the proposed system is its two-stage diagnostic workflow, where initial predictions are refined using an interactive questionnaire that simulates clinician–patient interaction. This combination of deep learning, explainability, clinical validation, and user interaction positions RetinAI as a comprehensive and practical decision-support tool for real-world retinal disease screening.

II. Literature Review

The modern era of deep learning for medical image analysis was fundamentally enabled by breakthroughs in general-purpose image recognition architectures developed on the ImageNet Large Scale Visual Recognition Challenge (ILSVRC), which comprises over 1.2 million training images across 1,000 categories. He et al. [1] introduced deep residual learning (ResNet) with the key innovation of skip (shortcut) connections that allow gradient flow to bypass one or more convolutional layers, thereby resolving the degradation problem wherein deeper networks paradoxically produce higher training error than their shallower counterparts. The proposed ResNet-152, consisting of 152 convolutional layers with approximately 60 million parameters, achieved a top-5 error rate of 3.57% on the ILSVRC-2015 benchmark — surpassing human-level performance for the first time in the competition's history. The residual learning framework is mathematically formulated as $H(x) = F(x) + x$, where $F(x)$ represents the learned residual mapping and x the identity shortcut, enabling the network to learn incremental refinements rather than complete transformations. This architecture has become a foundational backbone for transfer learning in medical imaging, as the pretrained residual features naturally capture hierarchical visual patterns transferable to pathological structures in retinal fundus images. Szegedy et al. [2] proposed the Inception architecture (GoogLeNet), which introduced the concept of multi-scale feature extraction within a single network layer through parallel convolutional branches of varying kernel sizes (1×1, 3×3, and 5×5 filters alongside 3×3 max-pooling), concatenated to form a rich feature representation. This "network within a network" design philosophy reduced the parameter count to approximately 6.8 million — roughly 12 times fewer than AlexNet — while achieving a top-5 error rate of 6.67% on ILSVRC-2014. The multi-scale approach is particularly advantageous for retinal image analysis, where pathological features span vastly different spatial scales: microaneurysms occupy only a few pixels, while optic disc changes and macular degeneration affect larger retinal

regions. Simonyan and Zisserman [3] demonstrated through VGGNet that architectural simplicity with uniform 3×3 convolutional filters across all layers, combined with increased network depth (16 and 19 weight layers in VGG-16 and VGG-19 respectively), could achieve state-of-the-art performance with a top-5 error rate of 7.3% on ILSVRC-2014. VGGNet's 138 million parameters make it computationally expensive; however, its uniform architecture and highly transferable intermediate features established it as the most widely adopted backbone for transfer learning in early medical imaging applications. The key insight that two stacked 3×3 convolutions provide the same effective receptive field as a single 5×5 filter while employing fewer parameters and introducing an additional non-linearity has influenced all subsequent architecture designs. Building upon the principle of feature reuse, Huang et al. [4] proposed DenseNet (Densely Connected Convolutional Networks), wherein each layer receives direct input from all preceding layers within a dense block via channel-wise concatenation, rather than through element-wise addition as in ResNet. DenseNet-121, comprising 121 layers organized into four dense blocks with growth rate k = 32, achieved 25.02% top-1 error on ImageNet while using only approximately 8 million parameters — substantially fewer than both ResNet-50 (25.5M) and VGG-16 (138M). This parameter efficiency arises because dense connections encourage feature reuse across layers, effectively reducing feature redundancy. For medical imaging tasks characterized by limited training data — typically fewer than 10,000 images in retinal disease datasets compared to millions in ImageNet — this parameter-efficient architecture is particularly advantageous, as it mitigates overfitting while maintaining expressive capacity. DenseNet's dense connectivity also facilitates implicit deep supervision, as earlier layers receive gradient signals directly from the loss function through shortened paths, enabling them to learn more discriminative low-level features such as retinal vessel edges and lesion boundaries. More recently, Tan and Le [5] introduced EfficientNet, which addresses the question of how to systematically scale CNN architectures for maximum accuracy per FLOP. Through a neural architecture search (NAS) process, they first designed a baseline network (EfficientNet-B0) and then proposed a compound scaling method that uniformly scales network width (number of channels), depth (number of layers), and input resolution using a compound coefficient φ, with scaling factors α^φ for depth, β^φ for width, and γ^φ for resolution, subject to the constraint $\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$. This principled scaling approach produced a family of models from EfficientNet-B0 (5.3M parameters, 77.1% top-1 accuracy) to EfficientNet-B7 (66M parameters, 84.3% top-1 accuracy), with each variant achieving superior accuracy-efficiency trade-offs compared to existing models. EfficientNet-B3, used in the proposed RetinAI ensemble system, comprises approximately 12 million parameters with an input resolution of 224×224, achieving 81.6% top-1 accuracy on ImageNet — making it a strong candidate for retinal image classification where computational resources and dataset sizes are limited.

The direct application of these foundational architectures to retinal disease screening was pioneered by the landmark study of Gulshan et al. [6], who developed and validated a deep learning algorithm specifically for the detection of diabetic retinopathy from retinal fundus photographs. The study employed an Inception-v3 architecture pretrained on ImageNet and fine-tuned on a large proprietary dataset of 128,175 retinal images, each graded 3–7 times by a panel of 54 US-licensed board-certified ophthalmologists to establish a high-quality reference standard. The algorithm was validated on two independent clinical datasets: EyePACS-1 (9,963 images from community clinics in the United States) and MESSIDOR-2 (1,748 images from three French ophthalmology departments), achieving an area under the receiver operating characteristic curve (AUC) of 0.991 on EyePACS-1 and 0.990 on MESSIDOR-2 at the operating point chosen for high sensitivity. At the selected threshold, the system achieved 97.5% sensitivity and 93.4% specificity for detecting referable diabetic retinopathy (moderate non-proliferative DR or worse), performance that exceeded all seven individual ophthalmologists on the validation panel. This study was transformative in establishing that deep learning could achieve ophthalmologist-level diagnostic accuracy for DR screening; however, the system was limited to binary classification (referable vs. non-referable DR), did not grade DR severity according to the ETDRS scale, could not detect co-existing conditions such as glaucoma or AMD, and relied on a proprietary training dataset that limits reproducibility. Building on this breakthrough, Ting et al. [7] developed and validated a deep learning system using retinal images from 76,370 patients drawn from the Singapore Integrated Diabetic Retinopathy Programme (SiDRP) and 10 multiethnic cohorts comprising 494,661 retinal images. The system employed a modified VGGNet architecture and was unique in its multi-disease scope, simultaneously screening for referable DR, vision-threatening DR, possible glaucoma, and AMD using independent binary classification heads. The reported AUCs were 0.936 for referable DR, 0.958 for vision-threatening DR, 0.942 for possible glaucoma suspect (based on vertical CDR ≥ 0.7 or neuroretinal rim narrowing), and 0.931 for AMD. The study was clinically significant for three reasons: first, it demonstrated that a single deep learning system could screen for multiple retinal diseases simultaneously; second, it validated performance across highly diverse populations including Chinese, Malay, Indian, Caucasian, and African American cohorts; and third, it operated within an established national screening infrastructure, suggesting practical clinical deployability. Nevertheless, the system treated each disease as an independent binary classification problem using separate softmax outputs, fundamentally precluding the modeling of disease co-occurrence patterns — for example, the well-established clinical association between diabetic retinopathy and hypertensive retinopathy in patients with metabolic syndrome. Gargeya and Leng [8] proposed an end-to-end data-driven deep learning approach for automated DR identification using a custom five-layer convolutional neural network architecture. The study utilized a training dataset of 75,137 color fundus images and validated performance on two independent external datasets: MESSIDOR-2 (1,748 images) and eOphtha (463 images). The system achieved an AUC of 0.97 on the primary dataset with 94% sensitivity and 98% specificity, and AUCs of 0.94 and 0.95 on the external validation sets respectively, using five-fold cross-validation for primary evaluation. A key contribution of this work was the generation of automated abnormality heatmaps —

data-driven attention maps highlighting regions that contributed most to the positive DR prediction. However, these heatmaps were coarse, unstructured visualizations without attribution to specific clinical features (exudates, hemorrhages, or microaneurysms), limiting their interpretive value for clinicians requiring structured diagnostic evidence. Abràmoff et al. [9] demonstrated improved automated DR detection through a hybrid approach that integrated deep learning features with handcrafted clinical descriptors, including hemorrhage detectors, exudate detectors, and texture features. The system was evaluated on the publicly available MESSIDOR-2 dataset (1,748 images from 874 diabetic patients), reporting an AUC of 0.980 with 96.8% sensitivity and 87.0% specificity at the reference standard operating point. The notable contribution of this study was its emphasis on evaluation using a publicly available, independently collected dataset — a critical step toward clinical deployment and reproducibility that many earlier studies lacked. Furthermore, this work was among the earliest to demonstrate that combining learned deep features with domain-specific handcrafted features could improve detection performance over either approach alone, presaging the hybrid methodology adopted by the proposed RetinAI system. Nonetheless, the validation dataset was relatively small (1,748 images), and the system was limited to binary DR detection without severity grading or multi-disease capability.

For disease-specific detection beyond DR, Li et al. [10] developed and validated a deep learning system specifically for detecting glaucomatous optic neuropathy (GON) based on color fundus photographs. Using a ResNet-50 architecture with transfer learning from ImageNet, the study employed a retrospective dataset of 48,116 fundus images from the Chinese community population and the LabelMe online annotation repository, labeled by a panel of trained graders with adjudication by glaucoma specialists. The system achieved an AUC of 0.986 with 95.6% sensitivity and 92.0% specificity on its primary validation set. The model was additionally validated on three external datasets — SCES (1,542 images), SINDI (4,095 images), and BES (487 images) — achieving AUCs of 0.977, 0.959, and 0.888 respectively, demonstrating strong but not uniform generalizability across different ethnic populations and imaging equipment. A critical limitation was the reliance on image-level glaucoma labels without explicit estimation of the cup-to-disc ratio (CDR), the most widely used clinical biomarker for glaucoma where a CDR exceeding 0.6 in Caucasian populations (or 0.7 in African populations) is considered suspicious. This disconnect between the model's classification mechanism and the clinician's diagnostic reasoning process exemplifies the broader challenge of clinical interpretability in deep learning systems. Orlando et al. [11] directly addressed the CDR estimation challenge through the REFUGE (Retinal Fundus Glaucoma Challenge), which established a unified benchmark framework for evaluating automated glaucoma assessment methods encompassing three subtasks: optic disc segmentation, optic cup segmentation, and binary glaucoma classification from fundus photographs. The challenge dataset comprised 1,200 images (400 training, 400 validation, 400 test) with pixel-level annotations for optic disc and optic cup boundaries. Twelve international teams submitted solutions employing diverse architectures including U-Net, DeepLabv3+, Attention-gated networks, and multi-task cascaded networks. The best-performing method for optic disc segmentation achieved a Dice coefficient of 0.960, while the best optic cup segmentation method achieved a Dice of 0.872. CDR estimation error was below 0.05 for the top methods, and the best glaucoma classification AUC reached 0.882. The REFUGE results demonstrated that deep segmentation-based CDR estimation can approach clinical-grade accuracy, providing a methodological foundation for incorporating spatial biomarkers into classification pipelines. However, the relatively small dataset (1,200 images) and the segmentation-only evaluation scope — without integration into a broader multi-disease classification framework — limited its direct applicability to comprehensive screening systems. Burlina et al. [12] applied deep convolutional neural networks to the automated grading of age-related macular degeneration from color fundus images. The study leveraged the Age-Related Eye Disease Study (AREDS) dataset comprising over 130,000 fundus images from 4,613 participants, with AMD severity graded according to the AREDS simplified scale. Two architectures were evaluated: OverFeat (a precursor to modern CNNs based on multi-scale sliding window detection) and VGGNet with transfer learning from ImageNet. The binary classification task distinguished between "no or early AMD" (AREDS categories 1–2) and "referable intermediate/advanced AMD" (AREDS categories 3–4), with the VGGNet-based model achieving an accuracy of 88.4% and a Cohen's kappa (κ) of 0.764, indicating substantial agreement with the reference standard defined by certified graders at reading centers. This κ value was comparable to inter-grader agreement among human experts, validating the clinical acceptability of the automated system. However, the binary formulation (referable vs. non-referable) precluded fine-grained staging of AMD severity, and the absence of any form of explainability — no heatmaps, no feature attribution, no clinical reasoning — raised concerns about clinical trust and regulatory approval.

The evolution toward multi-label retinal disease classification — wherein a single fundus image may simultaneously exhibit multiple co-existing pathologies — represents a critical paradigm shift from the binary and multi-class approaches described above. This transition is motivated by clinical reality: the ODIR-5K dataset reveals extensive disease co-occurrence, with diabetes and other abnormalities co-appearing in 497 of 7,439 samples, and hypertension co-occurring with diabetes in 89 samples. Wang et al. [13] proposed an EfficientNet-based multi-label classification system specifically designed for the ODIR-5K benchmark, which contains 7,439 retinal fundus images annotated across eight diagnostic categories — Normal (N), Diabetes (D), Glaucoma (G), Cataract (C), Age-related Macular Degeneration (A), Hypertension (H), Myopia (M), and Other Abnormalities (O). Their approach employed an ensemble of EfficientNet variants (B0 through B4) with sigmoid activation functions at the output layer for independent per-class probability estimation, replacing the softmax activation used in multi-class formulations. Binary cross-entropy loss was computed independently for each of the eight output neurons. The system incorporated test-time augmentation (horizontal flipping, multi-scale cropping) and model ensembling via probability averaging, achieving 98.7% accuracy, 97.7% precision, 98.7% recall, and 98.2%

F1-score on the ODIR competition test set. While these metrics are impressive, they should be interpreted with significant caution: the ODIR evaluation protocol and test/validation split configurations vary across studies, the dominant Normal and Diabetes classes (representing over 56% of all labels) inflate aggregate accuracy metrics, and the system provided no form of clinical explainability or feature validation. The absence of per-class metrics (e.g., per-class F1 for rare conditions like Hypertension with only 203 samples) makes it difficult to assess the system's viability for detecting underrepresented pathologies. Li et al. [14] introduced CANet (Cross-disease Attention Network), a novel architecture for joint diabetic retinopathy and diabetic macular edema grading that explicitly models inter-disease correlations. The architecture incorporates two novel attention modules: a Disease-Specific Attention Module (DSAM), which learns separate spatial attention maps for DR and DME to highlight disease-relevant regions, and a Disease-Dependent Attention Module (DDAM), which captures the mutual influence between DR and DME grades through cross-disease feature interaction. The network was built on a ResNet-50 backbone and evaluated on the IDRiD dataset (516 images with per-pixel annotations and image-level DR and DME grades) and MESSIDOR (1,200 images with DR grades). On IDRiD, CANet achieved quadratic weighted kappa values of 0.829 for DR grading and 0.886 for DME grading, representing statistically significant improvements over the baseline ResNet-50 without attention ($\kappa = 0.785$ for DR and $\kappa = 0.845$ for DME). The key contribution was demonstrating that explicitly modeling the clinical relationship between two co-occurring diseases — DR and DME frequently co-exist because DME is often a complication of DR — improves grading accuracy compared to independent classification heads that ignore this correlation. However, CANet's scope was limited to only two related conditions with an inherent clinical dependency; extending this attention-based correlation modeling to eight or more diverse conditions (some of which have no clinical correlation, e.g., cataract and myopia) remains an open research challenge.

The black-box nature of deep learning models presents a significant barrier to their clinical adoption in high-stakes medical decision-making. Selvaraju et al. [15] proposed Gradient-weighted Class Activation Mapping (Grad-CAM), a technique that generates coarse localization maps highlighting important regions in the input image for a specific CNN prediction. Unlike its predecessor CAM (Class Activation Mapping), which required the network to have a specific architecture with a global average pooling layer immediately before the output, Grad-CAM is applicable to any CNN architecture by computing the gradient of the target class score with respect to the feature maps of any convolutional layer, followed by global average pooling of these gradients to obtain importance weights per channel. The resulting heatmap is computed as a ReLU-activated weighted combination of forward activation maps, producing a coarse localization map at the spatial resolution of the chosen convolutional layer. Evaluated on VGG-16 and ResNet architectures across ImageNet classification and PASCAL VOC localization tasks, Grad-CAM improved class-discriminative localization accuracy by 10.4% over CAM and demonstrated that it could effectively distinguish between fine-grained categories based on the spatial distribution of attention. For medical imaging, Grad-CAM has become the de facto standard for CNN interpretability, enabling clinicians to verify whether model attention aligns with pathologically relevant regions such as exudates, hemorrhages, or the optic disc. However, the resolution of Grad-CAM visualizations is inherently limited by the spatial dimensions of the final convolutional layer's feature maps (typically 7×7 or 14×14 for 224×224 input), and the heatmaps provide no structured clinical information — they show "where" the model is looking but not "what" clinical feature it has identified. Rajpurkar et al. [16] applied DenseNet-121 to multi-label chest radiograph interpretation in a system named CheXNet, directly demonstrating the transferability of both the DenseNet architecture and the Grad-CAM explainability framework to medical multi-label classification. CheXNet was trained on the ChestX-ray14 dataset comprising 112,120 frontal-view chest X-ray images annotated with 14 pathology labels (including pneumonia, cardiomegaly, effusion, and nodule) using natural language processing of associated radiology reports. The DenseNet-121 backbone was pretrained on ImageNet, with the final classification layer replaced by a 14-neuron dense layer with sigmoid activation for multi-label prediction. The system achieved a mean AUC of 0.7632 across all 14 pathologies, with individual AUCs ranging from 0.506 for hernia to 0.895 for cardiomegaly. For pneumonia specifically, CheXNet achieved an AUC of 0.7680, exceeding the average radiologist performance (AUC 0.633 computed from four practicing radiologists). This study established two critical precedents for the proposed RetinAI system: first, that DenseNet-121 (one of the core architectures in our ensemble) with sigmoid outputs and binary cross-entropy loss is an effective and parameter-efficient architecture for multi-label medical image classification; and second, that Grad-CAM heatmaps overlaid on medical images provide clinically meaningful visual explanations that aid in diagnostic reasoning, even when classification performance is imperfect. Tjoa and Guan [17] provided a comprehensive survey on explainable artificial intelligence (XAI) in the context of medical applications, published in IEEE Transactions on Neural Networks and Learning Systems. The survey systematically categorized over 200 XAI methods into seven taxonomic families: (1) perturbation-based methods (e.g., LIME, SHAP) that assess feature importance by systematically occluding or modifying input regions; (2) backpropagation-based methods (e.g., Grad-CAM, Guided Backpropagation, DeconvNet) that use gradient signals to attribute predictions to input features; (3) attention mechanism-based methods that embed interpretability into the model architecture itself through learned attention weights; (4) concept-based methods that map internal representations to human-understandable concepts; (5) example-based methods that explain predictions by reference to similar training instances; (6) rule-based methods that distill complex model behavior into human-readable decision rules; and (7) intrinsically interpretable models such as decision trees and generalized additive models. The review argued that clinical trust in AI requires not merely accurate predictions but transparent reasoning, and identified three key requirements for medical XAI: fidelity (the explanation must accurately reflect the model's decision process), comprehensibility (the explanation must be understandable to clinicians without

machine learning expertise), and sufficiency (the explanation must provide enough information for the clinician to accept or override the prediction). This taxonomy directly informs the multi-modal explainability strategy of the proposed RetinAI system, which combines backpropagation-based visual explanations (Grad-CAM) with rule-based clinical validation. Sengupta et al. [18] conducted a critical review of deep learning techniques applied to fundus images for ophthalmic diagnosis, published in Artificial Intelligence in Medicine. The review analyzed over 100 published studies spanning DR detection, glaucoma screening, AMD classification, and retinal vessel segmentation. A central argument of the review was that while end-to-end deep learning systems achieve impressive aggregate classification metrics, they fail to provide the structured clinical features that ophthalmologists rely upon for diagnosis: the cup-to-disc ratio for glaucoma, the presence and count of microaneurysms for early DR detection, the presence of drusen for AMD staging, and the degree of retinal vessel attenuation for hypertensive retinopathy. The authors emphasized that clinically meaningful intermediate representations — not merely final classification scores — are essential for building physician trust and enabling clinical integration. This perspective directly motivates the proposed RetinAI system's hybrid architecture, which explicitly computes classical clinical biomarkers (exudate count, hemorrhage count, microaneurysm count, CDR, vessel visibility, texture variance, edge density) alongside CNN predictions and cross-validates them through literature-backed rules.

Ronneberger et al. [19] proposed U-Net for biomedical image segmentation, introducing an encoder-decoder architecture with symmetric skip connections that concatenate feature maps from the contracting path to the corresponding expansive path, thereby combining high-resolution spatial information from early layers with semantic context from deeper layers. U-Net was originally designed for neuronal structure segmentation in electron microscopy images (ISBI 2012 challenge dataset comprising only 30 training images with dimensions of 512×512 pixels), achieving an IoU (Intersection over Union) of 0.920 — winning the challenge by a significant margin. The architecture's most impactful innovation was its ability to achieve state-of-the-art segmentation performance with extremely limited training data through extensive data augmentation, including elastic deformations that simulate realistic biological tissue variations. In the domain of retinal image analysis, U-Net and its variants (Attention U-Net, U-Net++, and nnU-Net) have been widely adopted for three critical segmentation tasks: retinal blood vessel extraction from fundus images, optic disc and optic cup delineation for CDR computation, and lesion segmentation for DR grading. The segmentation-based approach to CDR estimation, as validated in the REFUGE Challenge [11], represents a clinically superior alternative to the brightness-thresholding approach currently employed in the proposed RetinAI system. An important methodological contribution addressing class imbalance — a pervasive and critical challenge in retinal disease datasets where the label distribution is severely skewed — is the focal loss proposed by Lin et al. [20]. In the ODIR-5K dataset, for example, Diabetes and Normal together constitute over 56% of all labels, while Hypertension represents only 2.7%, creating a 10:1 imbalance ratio between the most and least frequent classes. Standard binary cross-entropy loss treats all samples equally, causing the optimizer to be dominated by the loss contributions from the abundant easy-to-classify normal samples while under-learning the rare but clinically important pathological patterns. Focal loss addresses this by introducing a modulating factor $(1-p_t)^\gamma$ to the standard cross-entropy, where p_t is the model's estimated probability for the correct class and $\gamma \geq 0$ is a tunable focusing parameter. When $\gamma > 0$, the loss for well-classified examples ($p_t \to 1$) is down-weighted exponentially, directing the training gradient toward hard, misclassified examples that typically belong to rare pathological categories. With $\gamma = 2$ (the recommended setting) and an additional class-balancing factor $\alpha = 0.25$, focal loss achieved 39.1% AP on the COCO benchmark for dense object detection using the proposed RetinaNet architecture — a 3.2% improvement over standard cross-entropy. In the context of multi-label retinal disease classification, focal loss has been widely adopted to mitigate the detrimental effect of class imbalance on the detection of rare conditions such as hypertension, AMD, and other abnormalities; the proposed RetinAI system handles class imbalance by dynamically computing class-specific positive weights for a weighted standard binary cross-entropy loss.

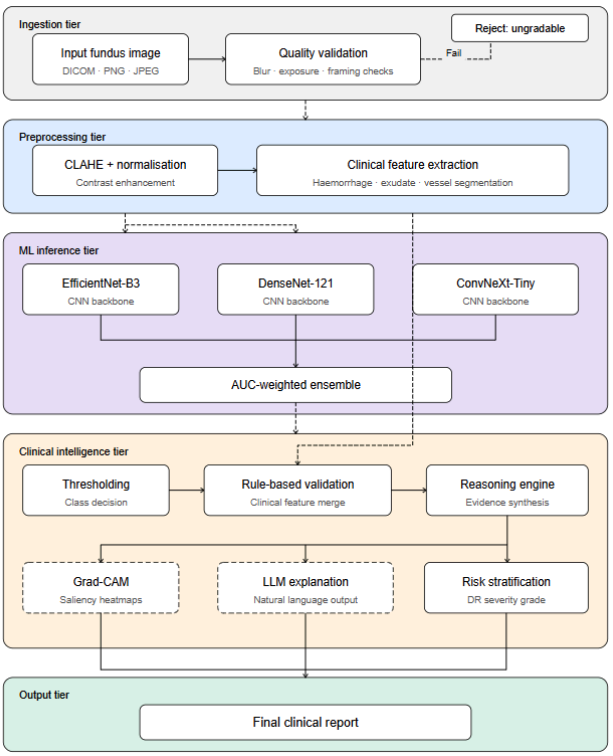
Despite these substantial advancements spanning foundational architectures, clinical validation studies, multi-label classification techniques, explainability methods, and class imbalance solutions, several critical research gaps persist that collectively motivate the proposed RetinAI system. First, the overwhelming majority of retinal AI systems — including the high-profile studies by Gulshan et al. [6], Ting et al. [7], and Li et al. [10] — operate as standalone CNN classifiers that produce disease probability scores without cross-referencing these predictions against classical clinical biomarkers such as exudate counts, hemorrhage patterns, or the cup-to-disc ratio. This practice departs fundamentally from the diagnostic reasoning process employed by ophthalmologists, who routinely integrate visual pattern recognition with quantitative clinical measurements and patient history. Second, multi-label detection on comprehensive datasets such as ODIR-5K remains a challenging open problem, with most systems achieving macro F1-scores below 0.50 due to severe class imbalance (10:1 ratio between dominant and rare classes) and the heterogeneous, often subtle visual presentation of rare conditions like hypertensive retinopathy and early AMD. The high aggregate accuracy metrics reported by studies such as Wang et al. [13] (98.7%) are dominated by performance on the majority classes and may obscure poor performance on clinically critical minority conditions. Third, explainability in existing systems is typically limited to post-hoc gradient-based heatmaps (Grad-CAM), which show where the model attends but provide no structured clinical reasoning, no literature-backed validation, and no integration with established clinical guidelines such as the ETDRS for DR severity grading or the AAO Preferred Practice Patterns for glaucoma management. Fourth, no existing system in the reviewed literature incorporates an interactive patient questionnaire to refine AI predictions based on symptom history and risk factors — a practice that is fundamental to clinical ophthalmology, where

patient-reported symptoms (blurred vision, metamorphopsia, floaters, eye pressure sensation) and medical history (diabetes duration, blood pressure status, family history of glaucoma, refractive error) significantly influence diagnostic confidence. The proposed RetinAI system addresses all four of these gaps by integrating an ensemble-based multi-label classification (EfficientNet-B3, DenseNet-121, ConvNeXt-Tiny) with classical computer vision feature extraction (exudates, hemorrhages, microaneurysms, CDR, texture variance, vessel visibility, edge density), a literature-backed rule-based clinical validation engine citing ETDRS Report No. 10, AAO Preferred Practice Patterns, Jonas ISNT Rule, Keith-Wagener-Barker Classification, and META-PM Study Group guidelines, Grad-CAM visual explainability with interactive comparison sliders, and a two-phase interactive diagnostic workflow that generates disease-specific clinical questions and contextualizes findings based on patient-reported answers (while strictly preserving the raw CNN confidence scores) — constituting a comprehensive hybrid clinical pipeline that more faithfully mirrors the multi-modal, multi-step diagnostic process of expert ophthalmologists than any existing system in the literature survey.

III. Methodology

The proposed RetinAI system is a hybrid clinical screening pipeline designed to generate structured and clinically interpretable retinal disease assessments from colour fundus images. The system integrates a multi-model deep learning ensemble with classical computer vision–based biomarker extraction, rule-based clinical validation, gradient-based visual explainability, and a patient-context interpretation module. The classification backbone consists of three complementary convolutional architectures—EfficientNet-B3, DenseNet-121, and ConvNeXt-Tiny—whose per-class sigmoid outputs are combined using a per-class AUC-weighted soft-voting strategy. This ensemble formulation leverages the distinct inductive biases of compound-scaled networks, densely connected feature reuse, and modernised convolutional designs to improve robustness and generalisation across heterogeneous retinal pathologies. Unlike conventional end-to-end systems that rely solely on a single neural network, the proposed approach explicitly incorporates clinically grounded reasoning layers that emulate the multi-modal diagnostic workflow of ophthalmologists. The system accepts a single retinal fundus image and processes it through a sequential eight-stage pipeline beginning with image quality validation to ensure reliability of downstream analysis. Following validation, the image undergoes CLAHE-based enhancement and standardised preprocessing before being passed simultaneously into two parallel branches: a deep learning inference branch and a classical feature extraction branch. In the inference branch, the preprocessed image is independently evaluated by the three backbone networks, producing multi-label sigmoid probability vectors that are fused via per-class AUC-weighted averaging. In parallel, the feature extraction branch computes a structured set of clinically relevant biomarkers—including lesion counts, vascular characteristics, and texture descriptors—directly from the raw image using deterministic computer vision techniques. The outputs of both branches are then integrated through a rule-based validation module that cross-references model predictions with extracted clinical evidence, followed by a reasoning engine that incorporates patient metadata such as age, systemic conditions, and reported symptoms to generate a risk-stratified clinical interpretation. A key architectural constraint enforced throughout the pipeline is that model-derived confidence scores remain immutable after inference. All downstream modules operate strictly as read-only consumers of the ensemble outputs, producing additive annotations, validation flags, and interpretive summaries without modifying the underlying probabilities. This design ensures statistical calibration is preserved while maintaining full auditability of the decision-making process. The final stages of the pipeline include Grad-CAM–based visual explainability for spatial localisation of disease-relevant regions and optional natural-language explanation generation for patient-facing communication. Collectively, this multi-stage design transforms raw model predictions into a comprehensive clinical decision-support output that combines statistical inference, visual evidence, and contextual reasoning within a unified framework

A. System Architecture



The RetinAI pipeline is structured as a sequential eight-stage processing framework implemented in Python 3.10 using PyTorch 2.x, timm, OpenCV 4.x, Albumentations, and NumPy. Fig. 1 illustrates the overall system workflow, beginning with the ingestion of a retinal fundus image and progressing through validation,

inference, and clinical reasoning stages. The pipeline first applies an image quality assessment module that evaluates four criteria: blur score using Laplacian variance with a threshold of 100.0, mean brightness constrained within the range 40.0–220.0, contrast measured via standard deviation with a threshold of 30.0, and optic disc detectability requiring at least 0.1% of the image area. If any critical condition—specifically excessive blur, insufficient brightness, or absence of the optic disc—is violated, the image is classified as ungradable and rejected with a structured diagnostic message. This early gating mechanism prevents unreliable inputs from propagating through the system and ensures downstream predictions remain clinically meaningful. Following successful validation, the pipeline bifurcates into two parallel branches that operate concurrently. In the first branch, the image undergoes CLAHE-based enhancement and ImageNet normalisation before being passed into the ensemble of three convolutional neural networks: EfficientNet-B3, DenseNet-121, and ConvNeXt-Tiny. Each model independently produces an eight-dimensional sigmoid probability vector corresponding to the ODIR-5K disease categories—Normal (N), Diabetes (D), Glaucoma (G), Cataract (C), Age-related Macular Degeneration (A), Hypertension (H), Myopia (M), and Other Abnormalities (O). These outputs are combined using a per-class AUC-weighted soft-voting mechanism, where each model's contribution to a given class is proportional to its validation AUC for that class. The fused probabilities are subsequently binarised using pre-optimised per-class thresholds and propagated unchanged to all downstream modules. In parallel, the second branch performs clinical feature extraction directly on the raw RGB image, generating a structured set of biomarkers including lesion counts, vascular measurements, texture descriptors, and quality indicators. The outputs from both branches converge in a rule-based clinical validation module that cross-references ensemble predictions against extracted features to generate supporting, contradicting, or informational clinical flags. These flags, along with the original ensemble probabilities and patient-provided contextual data such as age, diabetes status, hypertension status, and vision complaints, are then processed by a clinical reasoning engine. This module produces a structured diagnostic interpretation with risk stratification levels—routine, semi-urgent, or urgent—alongside evidence-backed explanations and referral recommendations. Additionally, an optional natural language generation component, implemented using Gemini 2.0 Flash or Groq LLaMA 3.3 70B as fallback, converts the structured output into patient-friendly explanations when API access is available, while a template-based fallback ensures robustness in offline scenarios. Finally, Grad-CAM heatmaps are generated for each condition exceeding its threshold, providing spatial visualisation of model attention over clinically relevant regions of the fundus image. Throughout this architecture, all post-inference components operate strictly as non-mutating layers, ensuring that original model confidence scores remain intact and preserving the integrity and auditability of the system.

B. Dataset and Preprocessing

This section describes the dataset characteristics, data partitioning strategy, preprocessing pipeline, augmentation scheme, and class imbalance handling techniques employed in the RetinAI system. The design of this stage ensures that the input data distribution is standardised, clinically meaningful variations are preserved, and class imbalance effects are mitigated prior to model training. Each component is carefully structured to maintain consistency across all three ensemble models while supporting robust generalisation under real-world imaging variability.

1) Dataset

The system is trained and evaluated on the Ocular Disease Intelligent Recognition (ODIR-5K) dataset, a multi-label ophthalmological benchmark comprising 7,439 retinal fundus images collected from 3,500 patients using diverse fundus imaging devices across multiple clinical environments. Each image is annotated with eight binary labels corresponding to Normal (N), Diabetes (D), Glaucoma (G), Cataract (C), Age-related Macular Degeneration (A), Hypertension (H), Myopia (M), and Other Abnormalities (O). The dataset inherently reflects real-world clinical conditions in which multiple pathologies may co-occur within a single patient, thereby requiring a multi-label classification framework rather than independent binary detection tasks. The distribution of labels is highly imbalanced, with Normal and Diabetes collectively accounting for over 56% of positive samples, while Hypertension represents only approximately 2.7% (around 203 samples), resulting in an approximate 10:1 imbalance ratio between the most and least frequent classes. This imbalance necessitates specialised training strategies such as class weighting and sampling adjustments. The dataset is selected due to its public availability, enabling reproducibility, its support for multi-disease annotation aligning with screening use cases, and its diversity in acquisition conditions, which enhances model robustness across varying clinical scenarios

2) Data Splitting

The dataset is partitioned into training and validation subsets using a multi-label stratified splitting strategy to preserve class distribution across all disease categories. When the iterative-stratification library is available, a MultilabelStratifiedKFold approach with $k = 5$and a fixed random seed of 42 is employed, from which a single fold is selected to produce an 80:20 split. This method ensures that rare classes such as Hypertension (approximately 2.7%) and Other Abnormalities (approximately 7.5%) maintain proportional representation in both training and validation sets, which is critical for stable evaluation. In scenarios where iterative stratification is unavailable, an approximate fallback strategy is implemented using scikit-learn's train_test_split, where stratification is performed based on the rarest positive label present in each sample while maintaining the same random seed for reproducibility. The resulting partition consists of approximately 5,951 training images and 1,488 validation images. Importantly, this identical split is used across all three ensemble models to ensure statistical consistency during validation and to enable reliable computation of ensemble weights and thresholds..

3) Preprocessing

Each retinal image undergoes a standardised preprocessing pipeline applied consistently during both training and inference, excluding augmentation during validation. First,

Contrast-Limited Adaptive Histogram Equalisation (CLAHE) is applied to enhance local contrast by converting the image from RGB to LAB colour space and operating on the luminance channel using a clip limit of 2.0 and a tile grid size of 8×8. This process improves the visibility of fine retinal structures such as blood vessels, exudates, and the optic disc while avoiding noise amplification. The image is then converted back to RGB and resized to a resolution of 224×224×3 pixels using bilinear interpolation, ensuring compatibility with all three backbone networks. Following resizing, pixel values are scaled to the range [0, 1] and standardised using ImageNet channel-wise statistics, aligning the input distribution with pretrained weights. The normalisation process is defined as:

$$x_{norm} = \frac{x / 255.0 - \mu}{\sigma}$$

where $\mu = [0.485, 0.456, 0.406]$and $\sigma = [0.229, 0.224, 0.225]$. Finally, the processed image is converted from a NumPy array of shape (224, 224, 3) to a channel-first tensor of shape (3, 224, 224) using Albumentations' `ToTensorV2`, and a batch dimension is appended during inference to produce the final input tensor of shape (1, 3, 224, 224). This pipeline ensures consistent feature representation across all models and stable convergence during training.

4) Data Augmentation

To improve generalisation and simulate real-world variability in fundus image acquisition, a comprehensive augmentation pipeline is applied during training using the Albumentations library. The augmentation process includes multiple transformations applied probabilistically to each image. First, random horizontal flipping with probability 0.5 simulates left–right eye symmetry, while vertical flipping with probability 0.2 accounts for occasional orientation variations. Illumination variability is introduced through brightness and contrast adjustments within ±0.15 at a probability of 0.5, and colour variability is simulated using hue (±10), saturation (±20), and value (±15) perturbations at a probability of 0.3. Geometric variability is incorporated through shift, scale, and rotation transformations with limits of ±5% shift, ±10% scaling, and ±15° rotation applied with probability 0.4. Additionally, Gaussian blur with kernel sizes ranging from 3 to 5 and probability 0.2 simulates minor focus imperfections, while coarse dropout with a maximum of four holes of size 20×20 pixels and probability 0.2 acts as a spatial regulariser by forcing the model to rely on distributed features. No augmentation is applied during validation or inference, where only deterministic preprocessing steps are used to ensure consistent evaluation.

5) Class Imbalance Handling

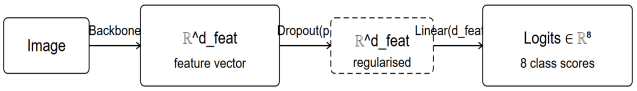
To address the severe class imbalance inherent in the dataset, two complementary strategies are employed during training. First, a per-class positive weighting scheme is applied within the loss function, where each class weight is computed as the ratio of negative to positive samples and clipped within the range [1.0, 20.0] to prevent instability. This weighting increases the contribution of underrepresented classes such as Hypertension and AMD during optimisation. Second, a WeightedRandomSampler is utilised to rebalance mini-batches by assigning sampling probabilities to each training instance based on the inverse frequency of its rarest positive label. This approach ensures that rare disease samples are more frequently encountered during training without discarding majority-class data, thereby preserving the overall data distribution while improving sensitivity to minority classes.

C. Model Architecture

This section describes the design of the multi-model ensemble classification backbone, including the architectural rationale, individual network configurations, ensemble fusion strategy, and threshold optimisation procedure. The architecture is specifically designed to balance representational diversity, statistical robustness, and class-wise adaptability, ensuring reliable performance across heterogeneous retinal disease categories.

1) Ensemble Design Rationale

The classification component employs a three-model ensemble rather than a single deep network to improve generalisation and reduce model-specific biases. This design is motivated by three key considerations. First, architectural diversity reduces correlated prediction errors, as different network families capture complementary feature representations; EfficientNet leverages compound scaling, DenseNet emphasises feature reuse through dense connectivity, and ConvNeXt incorporates modern convolutional design principles inspired by Vision Transformers. Second, ensemble averaging smooths the posterior probability distribution, thereby reducing overfitting to spurious correlations present in the relatively limited training dataset of approximately 6,000 images. Third, a per-class weighting mechanism enables the system to dynamically prioritise the model that performs best for a specific disease category, which is particularly important given the variability in classification difficulty across conditions. All three models follow a shared architectural template in which the input image is passed through a backbone feature extractor, followed by dropout regularisation and a fully connected layer producing an eight-dimensional logit vector, as formalised below:



2) EfficientNet-B3

EfficientNet-B3 is based on the compound scaling methodology, which uniformly scales network depth, width, and input resolution using a single scaling coefficient. The architecture utilises mobile inverted bottleneck convolution (MBConv) blocks combined with squeeze-and-excitation attention mechanisms to recalibrate channel-wise feature responses. With approximately 12 million parameters and a feature dimension of $d_{feat} = 1536$, it achieves a strong balance between computational efficiency and representational power. The classification head incorporates dropout with probability $p = 0.3$to mitigate overfitting during fine-tuning.

3) DenseNet-121

DenseNet-121 employs a dense connectivity pattern in which each layer receives feature maps from all preceding layers via concatenation. This structure facilitates feature reuse, strengthens gradient flow, and improves learning

efficiency for fine-grained visual patterns commonly present in retinal imagery. The model contains approximately 8 million parameters and produces a feature representation of dimension $d_{feat} = 1024$, making it the most lightweight component of the ensemble. Similar to EfficientNet-B3, the classification head uses dropout with probability $p = 0.3$to ensure regularisation.

4) ConvNeXt-Tiny

ConvNeXt-Tiny represents a modernised convolutional architecture that incorporates several design elements inspired by Vision Transformers, including larger kernel sizes (7×7), layer normalisation, GELU activation functions, and inverted bottleneck structures. With approximately 29 million parameters and a feature dimension of $d_{feat} = 768$, it provides the highest individual performance among the three models, achieving a validation AUC of 0.8628. Due to its larger capacity, a higher dropout probability of $p = 0.4$is used in the classification head to improve generalisation and reduce overfitting.

5) Per-Class AUC-Weighted Soft-Voting Ensemble

The outputs of the three models are combined using a per-class AUC-weighted soft-voting mechanism that assigns higher importance to models demonstrating superior discriminative performance for a given disease. For each class c, the weight assigned to model $m$is proportional to its validation AUC for that class and is normalised such that the weights sum to unity, as defined in equation (3). The final ensemble probability for each class is then computed as a weighted sum of individual model probabilities, as shown in equation (4):

$$w_{m,c} = \frac{AUC_{m,c}}{\sum_{j=1}^{3} AUC_{j,c}}$$

$$\hat{p}_c^{ensemble} = \sum_{m=1}^{3} w_{m,c} \cdot \hat{p}_{m,c}$$

where $\hat{p}_{m,c}$denotes the sigmoid probability predicted by model $m$for class c. This formulation enables adaptive weighting across disease categories, ensuring that the ensemble leverages the strengths of each model dynamically. The per-class weights used during deployment, derived from validation AUC scores, are presented below.

Class	EffNet-B3	DenseNet-121	ConvNeXt-Tiny
N	0.3288	0.3248	0.3463
D	0.3222	0.3274	0.3504
G	0.3265	0.3331	0.3404
C	0.3312	0.3334	0.3355
A	0.3212	0.3297	0.3492
H	0.3365	0.3286	0.3349
M	0.3353	0.3285	0.3362
O	0.3165	0.3201	0.3635

Table 1. Per-class AUC-weighted ensemble coefficients derived from validation performance.

The weight distribution indicates that ConvNeXt-Tiny consistently contributes the largest share across all classes, reflecting its superior overall AUC performance. The difference is particularly pronounced for the "Other Abnormalities" category, where its larger receptive field and modernised architecture provide improved capability in modelling heterogeneous pathologies.

6) Per-Class Optimal Thresholds

The ensemble probabilities are converted into binary predictions using per-class optimal thresholds determined through exhaustive grid search on the validation set. The search is conducted over the interval [0.05, 0.95] with a step size of 0.005, optimising the F1-score independently for each class. This approach accounts for differences in class prevalence, calibration, and decision boundary characteristics. The resulting thresholds are shown below.

Class	Optimal Threshold
N (Normal)	**0.27**
D (Diabetes)	**0.33**
G (Glaucoma)	**0.835**
C (Cataract)	**0.78**
A (AMD)	**0.725**
H (Hypertension)	**0.55**
M (Myopia)	**0.945**
O (Other)	**0.545**

Table 2. Per-class optimal decision thresholds obtained via validation F1-score maximisation.

The wide threshold range (0.270 for Normal to 0.945 for Myopia) reflects the model's varying calibration across classes and the class-specific trade-offs between sensitivity and specificity. The low threshold for Normal (0.270) ensures high sensitivity for detecting the absence of pathology, while the high threshold for Myopia (0.945) suppresses false positives in a condition where the model is highly confident when true disease is present.

D. Training Procedure

This section details the training methodology employed for all three ensemble models, including the transfer learning strategy, optimisation configuration, loss formulation, and stability mechanisms. The training process is designed to ensure efficient convergence, robustness to class imbalance, and reproducibility across runs while maintaining consistency across all backbone architectures.

1) Two-Phase Training Strategy

All three models are trained independently using a two-phase transfer learning protocol with mixed-precision computation (float16) on an NVIDIA Tesla T4 GPU with 16 GB VRAM, resulting in a total training time of approximately 111.2 minutes. In the first phase, referred to as frozen backbone warmup, the backbone parameters are kept non-trainable while only the classification head is optimised. This phase spans five epochs and enables the randomly initialised linear head to adapt to the ODIR-5K label space without perturbing pretrained ImageNet features through large gradient updates. The AdamW optimiser is used with an initial learning rate of 1×10^{-3}, and a cosine annealing schedule reduces the learning rate to 1×10^{-5}over the warmup duration. In the second phase, the entire network—including both backbone and classification head—is unfrozen and jointly fine-tuned for up to 35 additional epochs. During this stage, a reduced learning rate is applied to enable gradual adaptation of both low-level and high-level feature representations to retinal imagery, while a cosine annealing schedule further decreases the learning rate to 1×10^{-7}. Early stopping with a patience of 8 epochs monitors validation AUC and restores the best-performing model weights when no improvement is observed.

2) Per-Model Hyperparameters

The three backbone models employ slightly different hyperparameter configurations to account for differences in architecture size and capacity. EfficientNet-B3 and DenseNet-121 both use a fine-tuning learning rate of 3×10^{-5}, while ConvNeXt-Tiny uses a higher value of 5×10^{-5}due to its larger parameter space and modern design. Weight decay is set to $1\times10^{-5}$for EfficientNet-B3 and DenseNet-121, and $1\times10^{-4}$for ConvNeXt-Tiny to provide stronger regularisation. All models are trained for a maximum of 40 epochs, including 5 warmup epochs, and employ dropout probabilities consistent with their architectural definitions. The complete per-model configuration is presented below.

Hyperparameter	**EfficientNet-B3**	**DenseNet-121**	**ConvNeXt-Tiny**
timm model name	efficientnet_b3	densenet121	convnext_tiny
Feature dimension	1536	1024	768
Parameters (approx.)	12M	8M	29M
Dropout probability	0.3	0.3	0.4
Phase1 (warmup) LR	(1×10^{-3})	(1×10^{-3})	(1×10^{-3})
Phase2 (fine-tuning) LR	(3×10^{-5})	(3×10^{-5})	(5×10^{-5})
Weight decay	(1×10^{-5})	(1×10^{-5})	(1×10^{-4})
Phase 1 epochs	5	5	5
Total epochs	40	40	40
Best validation AUC	0.8234	0.8255	0.8628

Table 3. Per-model hyperparameter configuration for ensemble training.

3) Shared Training Configuration

In addition to model-specific parameters, a common training configuration is applied across all three models to ensure consistency and comparability of results. The input resolution is fixed at 224 × 224 × 3 pixels, and the output layer employs independent sigmoid activations for each of the eight classes to support multi-label classification. The AdamW optimiser is used with a batch size of 32, and training is conducted for a maximum of 40 epochs per model with early stopping based on validation AUC. Cosine annealing schedules are applied during both warmup and fine-tuning phases, while gradient clipping with a norm of 1.0 is enforced to stabilise updates. Mixed-precision training is enabled using automatic mixed precision (AMP) with GradScaler to improve computational efficiency. A fixed random seed of 42 ensures deterministic behaviour, and a 20% validation split is maintained using multi-label stratification. The ensemble combination strategy is consistent across all models, employing per-class AUC-weighted soft voting as described previously.

Hyperparameter	Value
Input resolution	224 × 224 × 3
Output activation	Sigmoid (per-class independent)
Number of classes	8
Optimizer	AdamW
Batch size	32
Total epochs(budget)	40 per model
Warmup epochs	5 per model
Early stopping patience	8 (monitor: val_auc, mode: max)
Phase 1 LR schedule	Cosine Annealing ((10^{-3} \rightarrow 10^{-5}))
Phase 2 LR schedule	Cosine Annealing (LR (\rightarrow 10^{-7}))
Gradient clipping norm	1.0
Loss function	BCEWithLogitsLoss (pos_weight)
Mixed precision	float16 (AMP via GradScaler)
Random seed	42
Validation split	20% (MultilabelStratifiedKFold)
Ensemble method	Per-Class AUC-Weighted Soft-Voting

Table 4. Shared training configuration across all ensemble models.

4) Loss Function

The training objective is defined using binary cross-entropy with logits, which integrates the sigmoid activation and cross-entropy computation into a numerically stable formulation. For a batch of Nsamples and $C = 8$classes, the loss is defined as:

$$BE = -\frac{1}{N\cdot C}\sum_{i=1}^{N}\sum_{c=1}^{C} w_c\left[y_{ic} + \left(1 - y_{ic}\right)\log log\left(1 - \sigma\left(z_{ic}\right)\right)\right]$$

where z_{ic} represents the raw logit for class c of sample i, $\sigma(\cdot)$ is the sigmoid function, $y_{ic} \in \{0, 1\}$ denotes the ground-truth label, and w_c is the class-specific positive weight. These weights are computed as:

$$w_c = clip\left(\frac{N_{neg,c}}{N_{pos,c}+\epsilon},\ 1.0,\ 20.0\right)$$

where $N_{pos,c}$ and $N_{neg,c}$ denote the number of positive and negative samples for class c, respectively, and $\epsilon = 10^{-6}$ prevents division by zero. Clipping the weights within the range [1.0, 20.0] ensures that extremely rare classes do not introduce instability during optimisation.

5) Training Stability Measures

Several mechanisms are incorporated to ensure stable and efficient training across all models. First, gradient clipping with a maximum norm of 1.0 is applied to prevent gradient explosion during fine-tuning. Second, mixed-precision training using float16 arithmetic via PyTorch's autocast and GradScaler reduces memory consumption and accelerates computation without compromising numerical stability. Third, batches producing non-finite loss values, such as NaN or infinite values, are automatically skipped to prevent corrupted updates. Finally, deterministic seeding is enforced through fixed random states and backend configurations, ensuring full reproducibility of results across multiple runs.

6) Ensemble Threshold Optimisation

After training, each model generates sigmoid probability predictions on the shared validation set, which are then combined using the per-class AUC-weighted ensemble strategy. The resulting ensemble probabilities are used to determine optimal decision thresholds for each class through exhaustive grid search over the interval [0.05, 0.95] with a step size of 0.005. The objective is to maximise the F1-score independently for each class, ensuring that threshold selection aligns with the final ensemble behaviour rather than individual model outputs. This post-training optimisation step is critical for balancing sensitivity and specificity across imbalanced disease categories and directly influences the final classification performance of the system

E. Clinical Feature Extraction and Rule-Based Clinical Validation

1) Clinical Feature Extraction Module

This module operates in parallel with the deep learning inference branch and is responsible for extracting interpretable, clinically relevant biomarkers directly from the raw RGB fundus image using classical computer vision techniques implemented in OpenCV. The primary objective of this component is twofold: first, to provide transparent and verifiable clinical evidence that complements model predictions, and second, to supply structured inputs for the downstream rule-based validation module. A total of 17 disease-specific features are computed across all eight ODIR categories, encompassing lesion detection, vascular morphology, texture characteristics, and global image quality indicators. These features are designed to approximate clinically observable signs used by ophthalmologists during manual diagnosis, thereby bridging the gap between black-box model outputs and interpretable medical reasoning.

a) Diabetic Retinopathy Features

Three key biomarkers are extracted for diabetic retinopathy detection. First, exudate count is computed using HSV colour space filtering with hue values between 15 and 40, saturation between 40 and 255, and value between 180 and 255, followed by morphological closing and opening operations and contour detection with a minimum area threshold of 30 pixels squared. Second, hemorrhage count is detected using a dual-range HSV filter targeting red hues in the ranges 0–10 and 160–180, combined with the same morphological processing pipeline to isolate candidate regions. Third, microaneurysm count is identified using OpenCV's SimpleBlobDetector, configured to detect small, dark, circular structures with area between 20 and 750 pixels squared, circularity greater than 0.5, convexity greater than 0.6, and inertia ratio above 0.4, applied on the CLAHE-enhanced green channel to maximise lesion contrast.

b) Glaucoma Features

The primary feature for glaucoma assessment is the cup-to-disc ratio (CDR), which is estimated using intensity-based segmentation. The optic disc is first identified by thresholding a Gaussian-blurred grayscale image at the 90th percentile of brightness, followed by morphological refinement using an 11×11 elliptical kernel with three closing and two opening iterations, and fitting a minimum enclosing circle. The optic cup is subsequently segmented using a higher threshold at the 95th percentile. The ratio of the cup radius to the disc radius is then computed and constrained within the range [0, 1], providing a clinically meaningful indicator of glaucomatous damage.

c) Myopia Features

Four features are extracted to characterise myopic changes. These include the mean brightness of the green channel, which reflects overall retinal illumination; texture variance computed from the grayscale image, capturing structural variability; vessel visibility measured as the mean absolute response of the Laplacian operator, indicating edge prominence; and edge density calculated as the proportion of pixels detected by the Canny edge detector with thresholds 50 and 150 relative to the total image area. Together, these features provide a quantitative representation of retinal texture and vascular clarity associated with myopic degeneration.

d) Cataract Features

Cataract-related features focus on image degradation caused by lens opacity. First, blur score is computed as the variance of the Laplacian of the grayscale image, where lower values indicate increased blur. Second, contrast ratio is measured as the standard deviation of grayscale pixel intensities, reflecting loss of contrast in hazy images. Third, haze index is defined as the ratio of the mean to the standard deviation of grayscale intensity, where higher values correspond to more uniform intensity distributions typical of cataract-affected images.

e) AMD Features

Age-related macular degeneration is characterised using features extracted from the central macular region, defined as the central third of the image. Drusen count is detected as bright spots using a 92nd-percentile intensity threshold, followed by morphological opening and filtering for circular regions with area between 10 and 500 pixels squared and circularity greater than 0.3. Additionally, macular irregularity is computed as the normalised standard deviation of the absolute Laplacian response within the macular region, capturing structural disruption associated with degeneration.

f) Hypertension Features

Two vascular features are computed to capture hypertensive changes. The arteriole-to-venule (AV) ratio proxy is derived from the vessel mask obtained via adaptive thresholding on the CLAHE-enhanced green channel, where vessel widths are estimated using a distance transform and the ratio of the 25th percentile to the 75th percentile of non-zero widths is calculated. Vessel tortuosity is quantified by computing the ratio of arc length to chord length for significant vessel contours exceeding 30 pixels in length and subtracting 1.0 to express deviation from straightness.

g) Normal and Other Features

For normal classification, global image clarity is quantified using Laplacian variance, and colour uniformity is measured as the mean of per-channel standard deviations. For other abnormalities, two features are computed: an anomaly score defined as the mean deviation of multi-orientation Gabor filter responses from their median, and irregularity density calculated as the fraction of pixels exceeding two standard deviations above the median response. These features capture heterogeneous patterns that do not fall into predefined disease categories.

2) Rule-Based Clinical Validation

The rule-based validation module integrates ensemble predictions with extracted clinical features to generate structured validation flags categorised as supporting, contradicting, or informational. This module is strictly non-invasive with respect to model outputs; it does not modify predicted probabilities but instead produces auxiliary annotations that enhance interpretability and clinical trust. The validation rules are derived from established ophthalmological guidelines, including ETDRS for diabetic retinopathy, AAO Preferred Practice Patterns for glaucoma, the ISNT rule, the Keith-Wagener-Barker classification for hypertensive retinopathy, the META-PM classification for pathological myopia, the AREDS and Beckman classifications for AMD, and the LOCS III system for cataract grading.

For each predicted disease, the module evaluates threshold-based conditions on the corresponding clinical biomarkers to determine consistency with established clinical criteria. For instance, a diabetes prediction accompanied by zero detected hemorrhages, exudates, and microaneurysms results in a high-severity contradicting flag, indicating inconsistency with expected pathological signs. Conversely, a glaucoma prediction with a cup-to-disc ratio exceeding 0.8 generates a high-severity supporting flag, reinforcing the model's prediction. In addition to single-disease validation, cross-disease interactions are also considered; for example, simultaneous predictions of diabetic retinopathy and hypertension trigger an informational flag highlighting shared vascular manifestations and known clinical co-morbidity. Similarly, concurrent predictions of Normal and any disease category produce a high-severity contradiction due to their mutual exclusivity within the ODIR grading framework. These rule-based validations provide an additional layer of clinical reasoning that complements statistical model outputs, enabling more reliable and interpretable diagnostic assessments

F. Evaluation Metrics and Explainability

1) Evaluation Metrics

This section outlines the quantitative metrics used to evaluate model performance on the held-out validation set, ensuring a comprehensive assessment of multi-label classification behaviour. Given the imbalanced and multi-label nature of the ODIR-5K dataset, multiple complementary metrics are employed to capture different aspects of predictive performance. These metrics are computed at both macro level, where each class contributes equally, and micro level, where predictions are aggregated across all classes. This dual evaluation strategy ensures that performance on rare conditions is not overshadowed by dominant classes while also reflecting overall system effectiveness.

Precision is defined as the proportion of predicted positive instances that are true positives, as given by equation (7). Recall, also referred to as sensitivity, measures the proportion of actual positive instances that are correctly identified, as shown in equation (8). The F1-score combines precision and recall into a single harmonic mean, balancing false positives and false negatives, as expressed in equation (9):

$$Precision = \frac{TP}{TP+FP}$$

$$Recall = \frac{TP}{TP+FN}$$

$$F_1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

To provide a threshold-independent evaluation, the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) is used as the primary metric for training, model selection, and ensemble weighting. AUC measures the model's ability to distinguish between positive and negative classes across all possible decision thresholds and is defined as:

$$AUC = \int_0^1 TPR(t)\, d(FPR(t))$$

where the true positive rate $TPR(t)$and false positive rate $FPR(t)$vary as functions of the classification threshold t. This metric is particularly suitable for imbalanced datasets, as it remains insensitive to class distribution. Additionally, Hamming Loss is employed to quantify the fraction of incorrectly predicted labels across all samples and classes, capturing both false positives and false negatives in a multi-label setting:

$$H = \frac{1}{N \cdot C} \sum_{i=1}^{N} \sum_{c=1}^{C} 1\left[\hat{y}_{ic} \neq y_{ic}\right]$$

where $1[\cdot]$is the indicator function. Unlike strict accuracy, Hamming Loss evaluates errors at the individual label level, making it more appropriate for multi-label classification tasks. Furthermore, class-wise 2×2 confusion matrices are computed independently for each disease category, enabling detailed analysis of true positives, false positives, true negatives, and false negatives, which is particularly valuable for identifying weaknesses in rare classes. Using the optimised ensemble and per-class thresholds, the system achieves a macro AUC-ROC of 0.8600, macro F1-score of 0.6280, micro F1-score of 0.5962, macro precision of 0.6192, macro recall of 0.6774, and a Hamming Loss of 0.1439.

2) Grad-CAM Visual Explainability

To enhance interpretability and clinical trust, Gradient-weighted Class Activation Mapping (Grad-CAM) is employed to generate spatial heatmaps that highlight regions of the fundus image contributing most strongly to each predicted disease class. The implementation targets the final convolutional layer of the EfficientNet-B3 backbone, which serves as the primary explainability model due to its balanced performance and stable feature representations. For a given class c, the gradient of the corresponding logit $z_c$with respect to the feature maps $A^k$is computed via backpropagation, and channel-wise importance weights are obtained using global average pooling, as defined in equation (12):

$$\alpha_k^c = \frac{1}{H \times W} \sum_i \sum_j \frac{\partial z_c}{\partial A_{ij}^k}$$

These weights are then used to compute the Grad-CAM heatmap as a weighted combination of feature maps followed by a rectified linear activation, as shown in equation (13):

$$L_{Grad-CAM}^c = ReLU\left(\sum_k \alpha_k^c A^k\right)$$

The resulting heatmap is normalised to the range [0, 1], resized to match the input resolution of 224×224 pixels, and mapped to a JET colour scheme. It is subsequently overlaid on the original fundus image using alpha blending with weights of 0.6 for the original image and 0.4 for the heatmap, producing an interpretable visual representation of model attention. Heatmaps are generated only for disease classes whose predicted probabilities exceed their respective optimal thresholds, ensuring that visual explanations correspond to clinically relevant predictions. This mechanism allows clinicians to verify whether the model focuses on anatomically meaningful regions such as the optic disc in glaucoma, the macular region in AMD, or vascular structures in diabetic and hypertensive retinopathy, thereby improving transparency and facilitating clinical validation of AI-generated outputs.

G. Clinical Reasoning, Patient Context, and Deployment

1) Clinical Reasoning and Risk Stratification

This module synthesises ensemble predictions, rule-based validation outputs, and patient-specific contextual data to generate a structured and clinically interpretable assessment. The objective is to transform probabilistic outputs into actionable diagnostic insights through risk stratification and evidence-based reasoning. Risk levels are determined relative to per-class optimal thresholds with an additional safety margin, ensuring consistent interpretation across disease categories. The risk formulation is defined as:

Risk Level= { High RiskHigh RiskBorderlineLow Riskif p≥0.80if p≥θc+0.15if p≥max(θc−0.15,0)otherwise

where $p$denotes the ensemble probability and $\theta_c$represents the class-specific optimal threshold. The inclusion of an absolute safety floor at $p \geq 0.80$ensures that highly confident predictions are always classified as high risk, even for diseases with elevated thresholds such as glaucoma and myopia. Additionally, an entropy-based out-of-distribution (OOD) detection mechanism is incorporated, flagging predictions as uncertain when the mean binary entropy across all classes exceeds 0.65, indicating a lack of confident model consensus. This reasoning framework ensures that diagnostic outputs are not only statistically grounded but also clinically interpretable and safe for downstream decision-making.

2) Patient Context Module and Symptom Stratification

The patient context module augments model predictions with structured demographic and symptomatic information to provide personalised and clinically meaningful recommendations. Inputs include patient age, diabetes status, hypertension status, and self-reported vision concerns expressed in simplified, patient-friendly language such as blurry vision, peripheral vision loss, or visual disturbances like floaters. These inputs are cross-referenced with model predictions and clinical findings to generate contextual annotations and risk modifiers. Symptom consistency is evaluated using a three-tier stratification scheme in which findings are categorised as consistent match when symptoms fully align with detected conditions, partial match when only some symptoms are explained, and minimal or review-required when symptoms are present without corresponding model-detected pathology. These categories are presented through intuitive visual indicators and integrated into the final clinical report. Importantly, all contextual interpretations remain strictly additive and do not influence the underlying ensemble predictions or confidence values. Risk modifiers are derived from established clinical evidence; for example, a confirmed diabetes history increases the risk weighting for diabetic retinopathy based on ETDRS findings, while age greater than 40 elevates glaucoma risk according to AAO guidelines and epidemiological studies. Conversely, absence of systemic conditions may reduce associated disease likelihood. This module enables the generation of tailored recommendations such as routine diabetic screening intervals or escalation to advanced imaging modalities when risk factors are present, thereby enhancing the clinical relevance of the system's output.

3) System Deployment and Clinical Reporting

The RetinAI system is deployed as a web-based clinical decision support platform using the FastAPI framework, providing a RESTful backend integrated with Jinja2-based frontend interfaces for both patients and clinicians. The backend architecture is organised into distinct route groups handling patient interactions, clinician workflows, and authentication processes. The trained ensemble models are dynamically loaded from a configuration file specifying architecture types, checkpoint paths, dropout parameters, ensemble weights, and threshold values, and are cached in memory to ensure efficient inference. In scenarios where ensemble loading fails, the system defaults to a single EfficientNet-B3 model to maintain operational continuity.
The system generates a comprehensive clinical screening report in PDF format using the ReportLab library, structured into ten sections including a risk assessment summary, key findings table with probability scores, detailed clinical interpretation, and symptom assessment mapping. Grad-CAM visualisations are embedded within the report to provide spatial evidence supporting each prediction, and all generated artefacts are stored and served through a structured file system integrated with a SQLite database for metadata tracking. The pipeline supports both CPU and GPU execution, with automatic mixed-precision enabled when GPU resources are available. The entire workflow—from image upload to report generation—is executed within a single request cycle, enabling real-time deployment in clinical and telemedicine environments.

In summary, the RetinAI methodology presents a comprehensive hybrid screening framework that integrates a per-class AUC-weighted ensemble of EfficientNet-B3, DenseNet-121, and ConvNeXt-Tiny with classical computer vision–based biomarker extraction, literature-driven rule-based validation, and context-aware clinical reasoning. The system incorporates threshold-optimised multi-label classification, entropy-based uncertainty detection, Grad-CAM visual explainability, and patient-specific contextual interpretation to produce structured, evidence-backed diagnostic outputs. Each stage of the pipeline is explicitly designed to preserve the statistical integrity of model predictions while progressively enriching them with clinically interpretable information and actionable insights. This layered architecture enables the system to closely emulate the multi-modal diagnostic reasoning process of expert ophthalmologists, thereby enhancing both reliability and practical applicability in real-world retinal disease screening scenarios

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